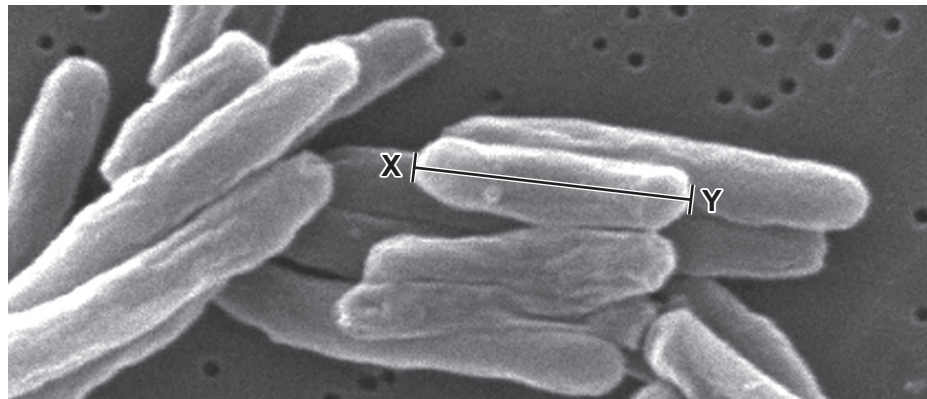


4 Tuberculosis (TB) is a major cause of ill health worldwide.

(a) State the name of a bacterium that causes TB in humans.

..... [1]

(b) Fig. 4.1 is a scanning electron micrograph of bacteria that cause TB.



magnification $\times 21\,000$

Fig. 4.1

Calculate the actual length of the bacterial cell shown in Fig. 4.1, along the line **X–Y**.

Write the formula you will use in the box.

Give your answer in micrometres (μm) to two significant figures.

formula

actual length = μm
[2]

(c) Bacteria are unicellular prokaryotic cells with a diameter of $1\text{--}5\,\mu\text{m}$.

State **two other** structural features that would identify a cell as prokaryotic.

1

2

[2]

- (d) The World Health Organization (WHO) Global Tuberculosis Report for 2019 published data on the estimated number of deaths from TB and HIV/AIDS in 2018. All deaths of people from TB who were infected with HIV were also counted as deaths of people with HIV/AIDS.

Fig. 4.2 shows these data. The dark grey boxes show the estimated number of deaths of people from TB who were also counted as deaths of people with HIV/AIDS.

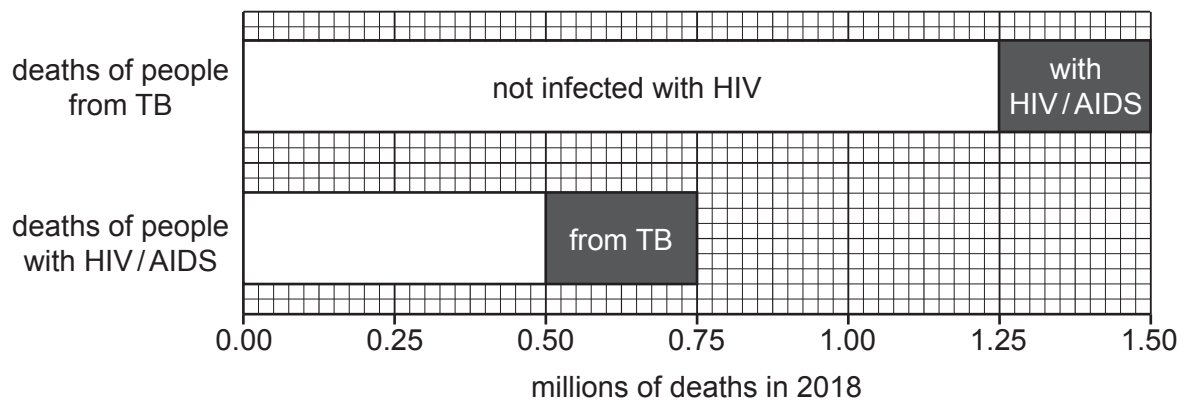


Fig. 4.2

A student used the data in Fig. 4.2 to predict that measures to control the spread of HIV will decrease the number of deaths from TB.

Discuss whether the data in Fig. 4.2 support this prediction.

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..... [3]

- (e) In healthy people, the number of T-helper cells ranges from 500 to 1200 cells per cm^3 of blood. In untreated people infected with HIV, the number of T-helper cells can decrease to below 200 cells per cm^3 of blood.

Explain how a low number of T-helper cells makes it more likely that untreated people infected with HIV will die if they are also infected with TB.

.....

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.....

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..... [3]

[Total: 11]